Dear Dr. Thrall,

Please consider our paper, "Traits predict forest phenological responses to photoperiod more than temperature" for publication as a letter in *Ecology Letters*.

Climate change is altering the environmental cues of growth globally, producing shifts in the timing of life history events (phenology), altering resource use and producing novel species interactions. In plant communities, predicting these shifts is critical to predicting changes in ecosystem services, like carbon sequestration. But to date, this has proven difficult because phenology is rarely included in studies of growth strategies[1] and is not included in frameworks like the leaf or wood economic spectra[2,3]. Yet phenology is foundational to the theory that underlies these spectra[4], which clearly predicts early to late phenology should align with acquisitive to conservative growth strategies and relating traits. This prediction has rarely been tested because phenology is often highly variable when measured in natural conditions, however, experiments and new models may overcome this challenge.

Here, we paired two massive controlled environment experiments with trait data for over 1400 individual plants across 47 species and eight sites, testing how phenology fits within major frameworks for plant growth. Based on decades of research into the relationship between traits and plant growth strategies, we explored whether early versus late leafout is correlated with acquisitive versus conservative growth strategies, and how these relationships change between forests across North America.

We found that phenology fits within major frameworks for plant growth, with early to late budburst correlating with size and leaf metrics. Of the five traits we measured, four traits related to leafout through photoperiod cues. Despite previous studies finding temperature to be the dominant controller of leafout, we found no support that traits relationships with temperature cues influence leafout timing. These results emerge in addition to trait variation across sites, with most traits varying across our latitudinal gradients and/or differing between our eastern and western transects of North America.

Our results provide the most compelling evidence of trait-phenology relationships with implications for how woody plant species will shift with climate change. Photoperiod is generally considered the weakest cue of leafout, but its shared importance with other species traits and shaping growth strategies suggest it may pose a major constraint on how species adapt to warmer climates than previous thought. Accounting for these phenology-trait relationships can improve forecasts of forest ecosystem and community change with warming, but have larger implications for how we understand phenology and growth strategies more broadly.

All authors contributed to this work and approve this version for submission. The manuscript is 4326 words with a 154 word summary, and 4 figures and is not under consideration elsewhere. We hope you find it suitable for publication in *Ecology Letters*, and look forward to your response.

Sincerely,
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